Preliminary Architecture

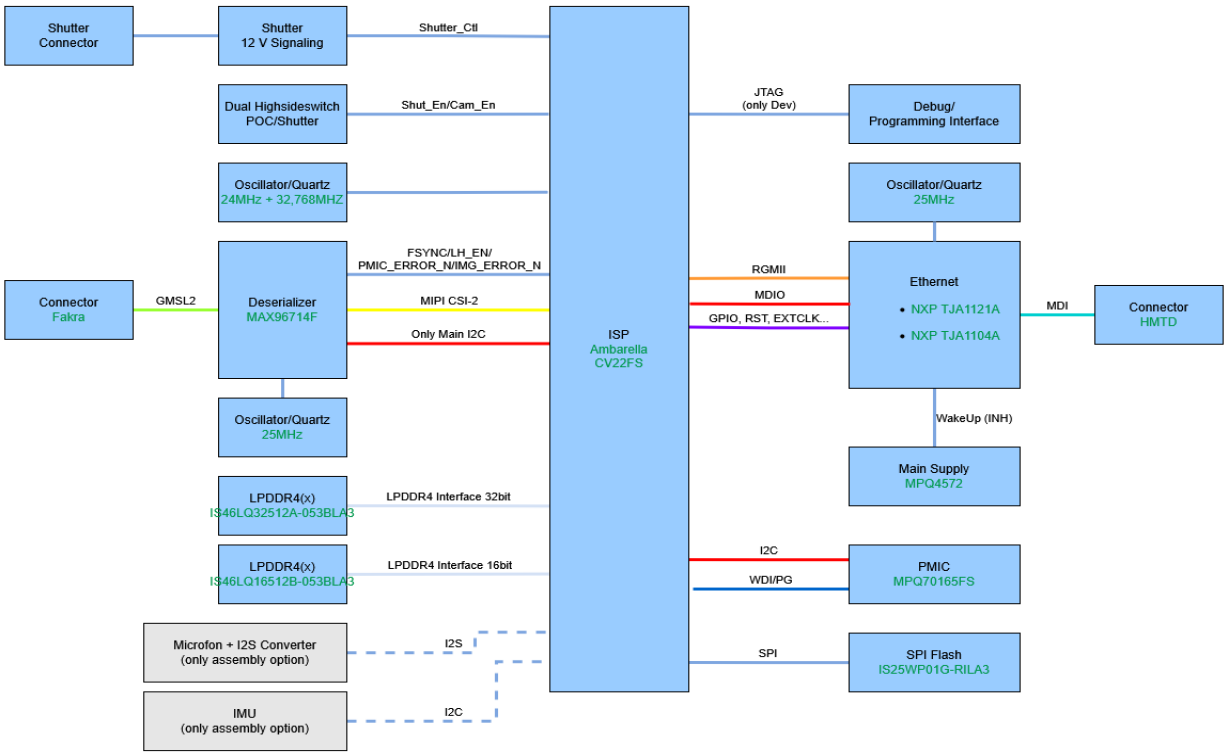
1 Document Scope 1
2 ECU Architecture 1
3 Satalite Camera Architecture 3
4 Overview of Draft Architecture Diagrams 4
4.1 Ethernet 4
4.2 Heater 5
4.3 Shutter 6
4.4 Video Streaming 7
4.5 PTP 14
4.6 Camera Interface 16
4.7 State Handling 16

1 Document Scope

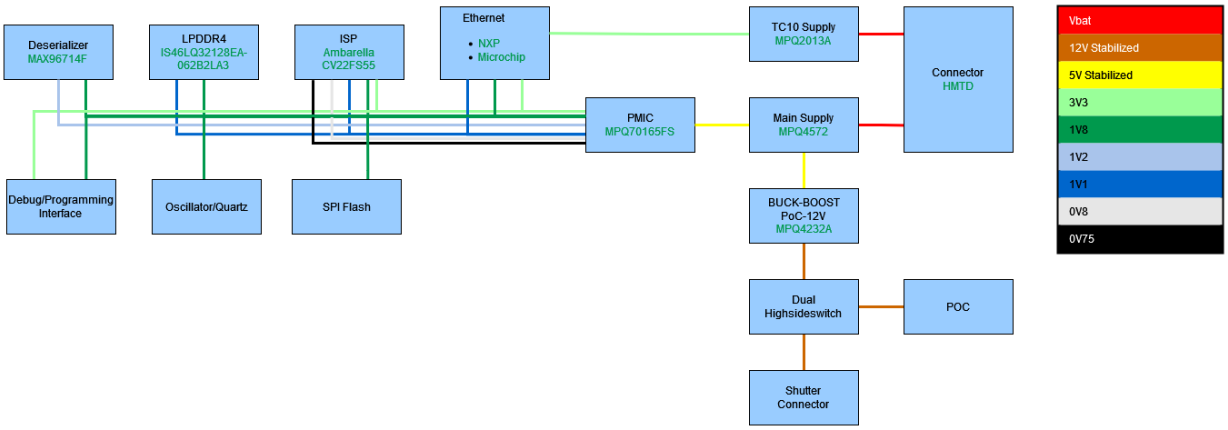
This document shall be used to document the preliminary architecture of the SOW2 ECU with its different components and interfaces.

2 ECU Architecture

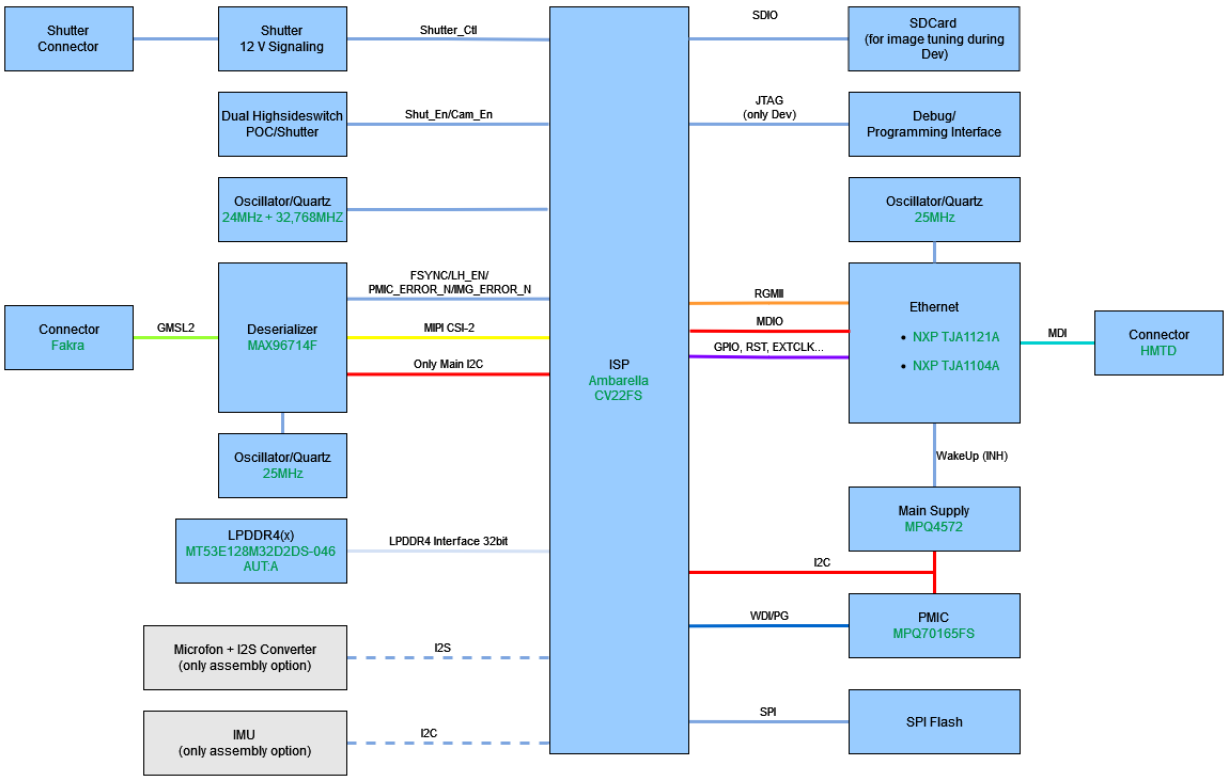
MVC-90506 - Preliminary SOW2 Architecture: Communication



MVC-90505 - Preliminary SOW2 Architecture: Power Management

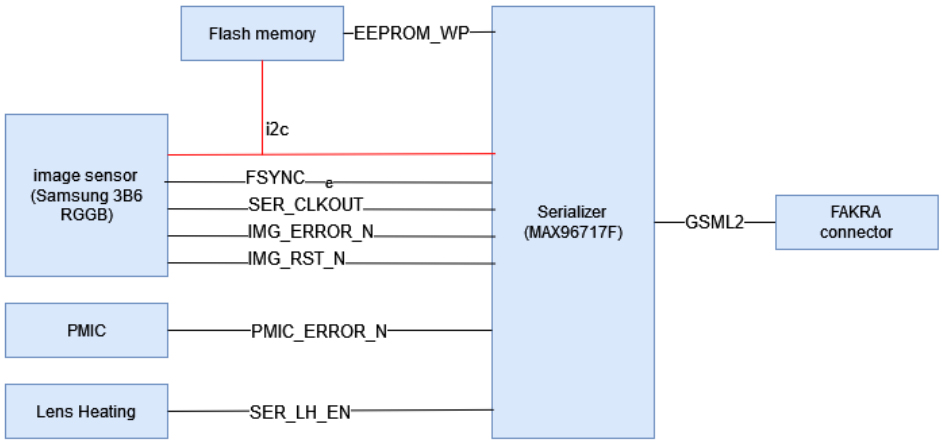


MVC-93033 - Preliminary Architecture - October 2024



3 Satalite Camera Architecture

MVC-92909 - Preliminary SOW1 Architecture: communication



MFP	Name of net	Description of net	External Pulls	Default state (camera powered, no config of camera)	Target Configuration of MFP
0	IMG_RST_N	Enable (high) or disable (low) imager	Pull down	Low	Push-pull high or Low
1	CFG0_FSYNC	Strap during start-up; Frame Sync connection to imager	Strap set to be I2C, XTAL, 0x80		Push-pull high or Low
2	CFG1	Strap during start-up;	Strap set to be COAX, Pixel, 3Gbps		
3	SER_LH_EN	Enable (high) or disable (low) lens heating	Pull down	Low	**Push-pull high or Low**
4	SER_CLKOUT	Programmable clock output of the serializer; Used as imager clock input		Low	25MHz
5	PMIC_ERROR_N	Error input from PMIC; No error (high), error (low)		High or low	Input internal pull down
6	VARIANT_CODE	Voltage divider input: low: B1-Sample (w/ LH) high: B0-Sample (wo/ LH)	**Pull down**	**Low**	Input No internal pull
7	IMG_ERROR_N	Error input from imager; No error (high), error (low)	Pull down	High or low	Input internal pull down
8	EEPROM_WP	Write protect of EEPROM: high: protected low: write enabled	Pull up	High	Push-pull high or Low

MIPI configuration in camera module

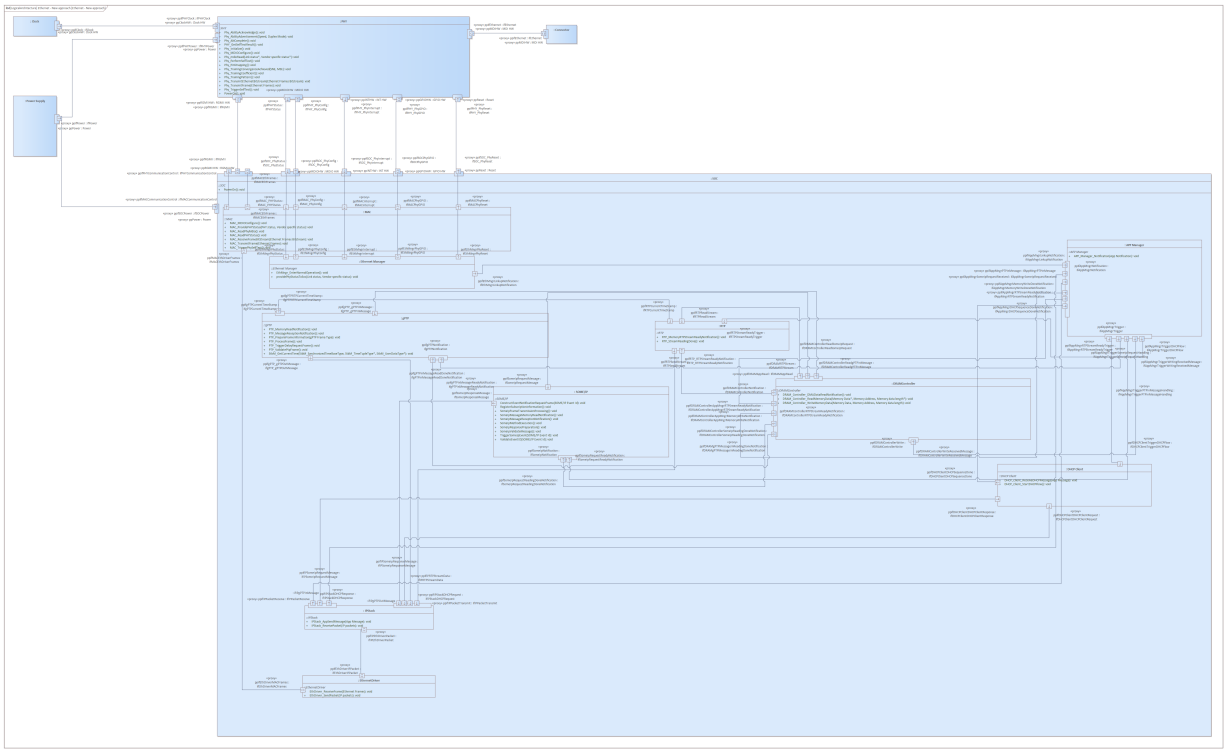
Image Sensor Output		Serializer Input	
Lane	Polarity	Lane	Polarity
CLK	N	CLK	P
	P		N
0	N	1	P
	P		N
1	N	0	P
	P		N
2	N	2	P
	P		N
3	N	3	N
	P		P

Register setting of Serializer to accommodate lane and polarity swaps:
[Slave Address (8-Bit), Register Address (16-Bit), Register Value (8-Bit)]
0x80, 0x0332, 0xE0
0x80, 0x0333, 0x01
0x80, 0x0334 , 0x10
0x80, 0x0335 , 0x07

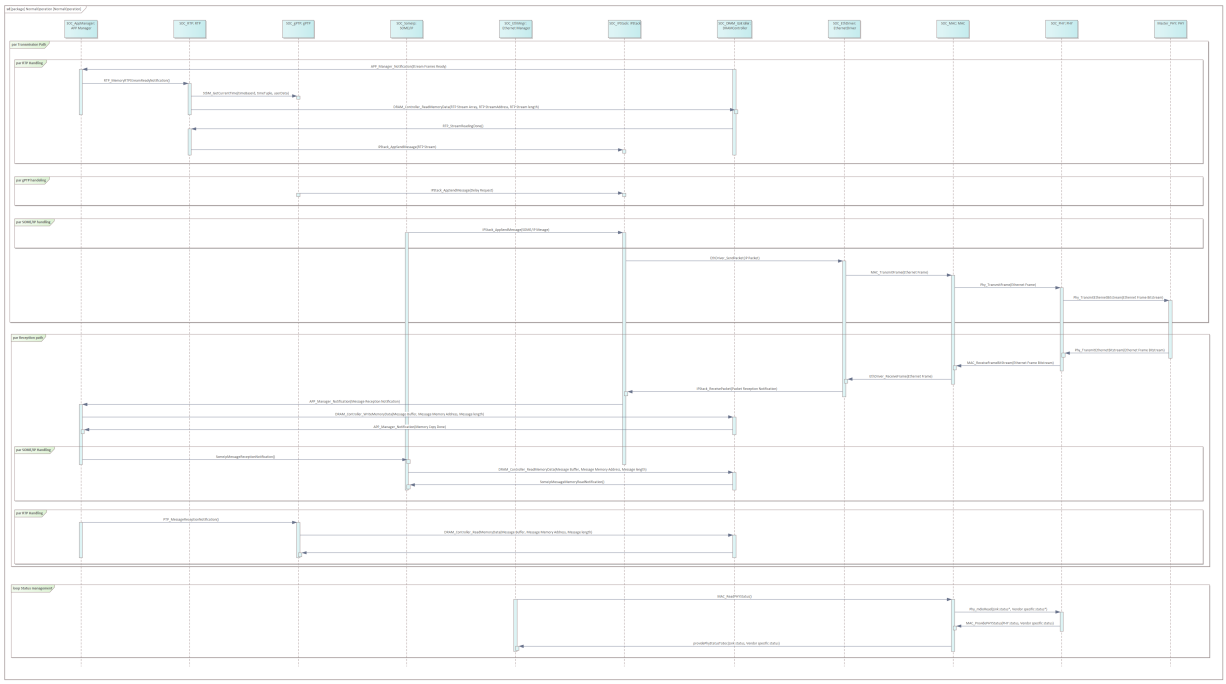
4 Overview of Draft Architecture Diagrams

4.1 Ethernet

MVC-100993 - Structural

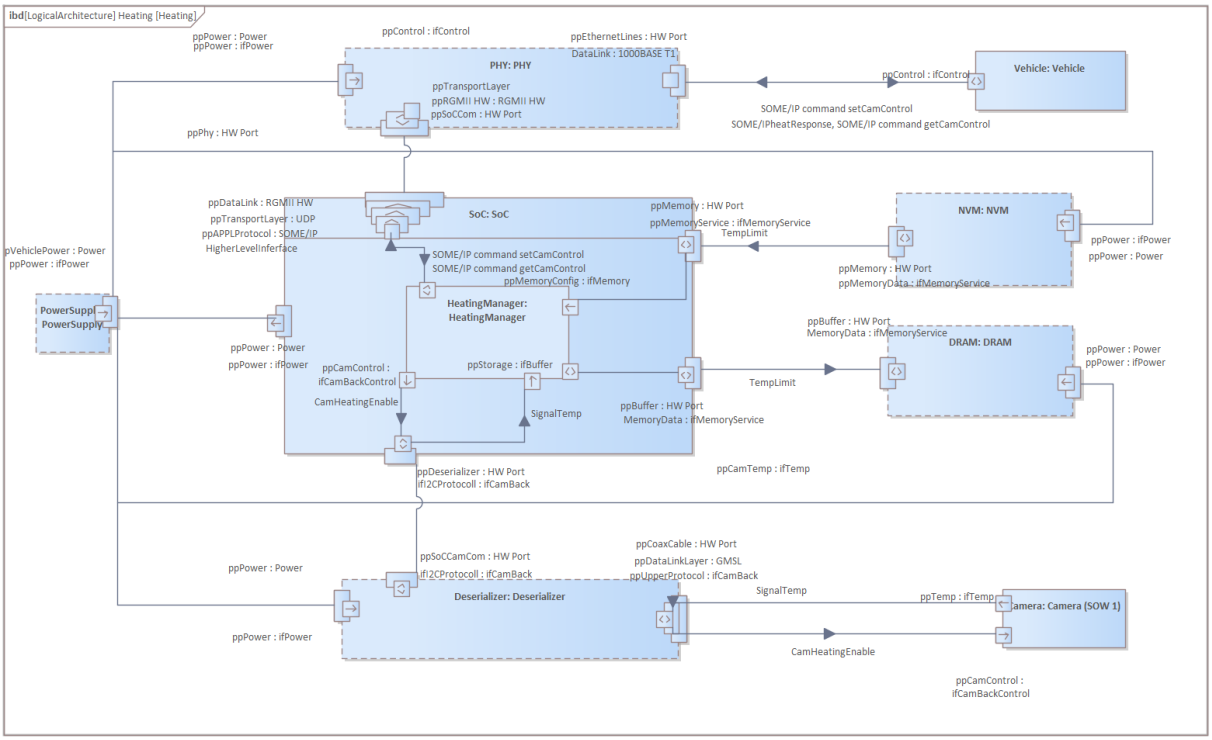


MVC-101008 - Behavioral

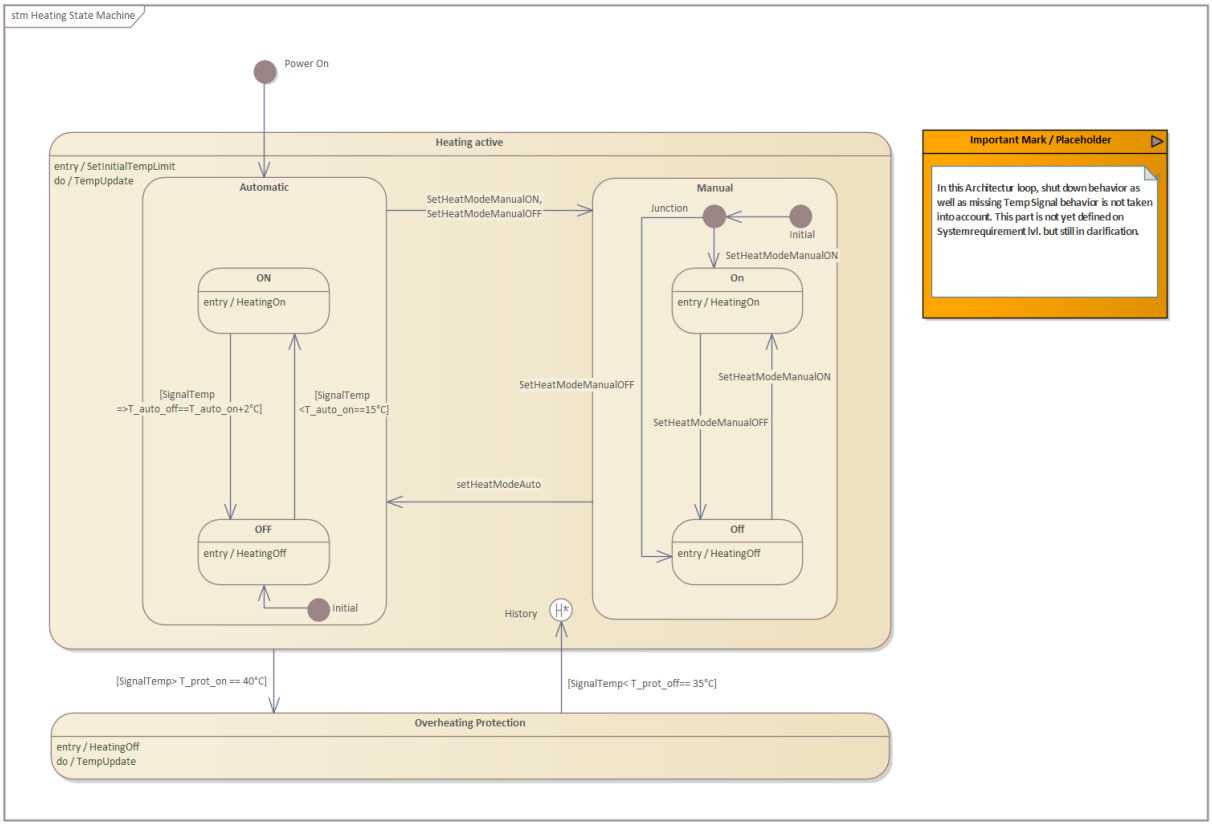


4.2 Heater

MVC-100994 - Structural

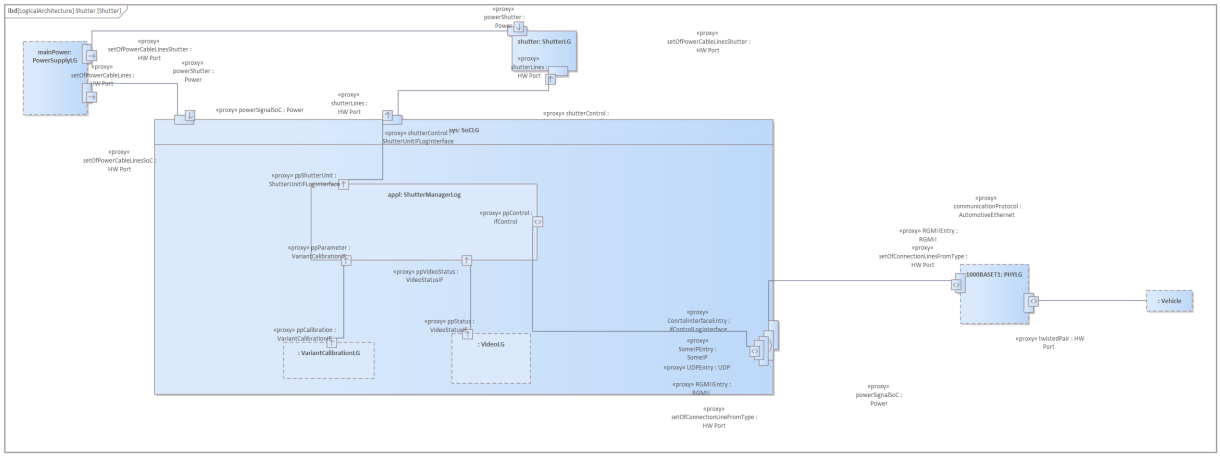


MVC-101009 - Behavioral

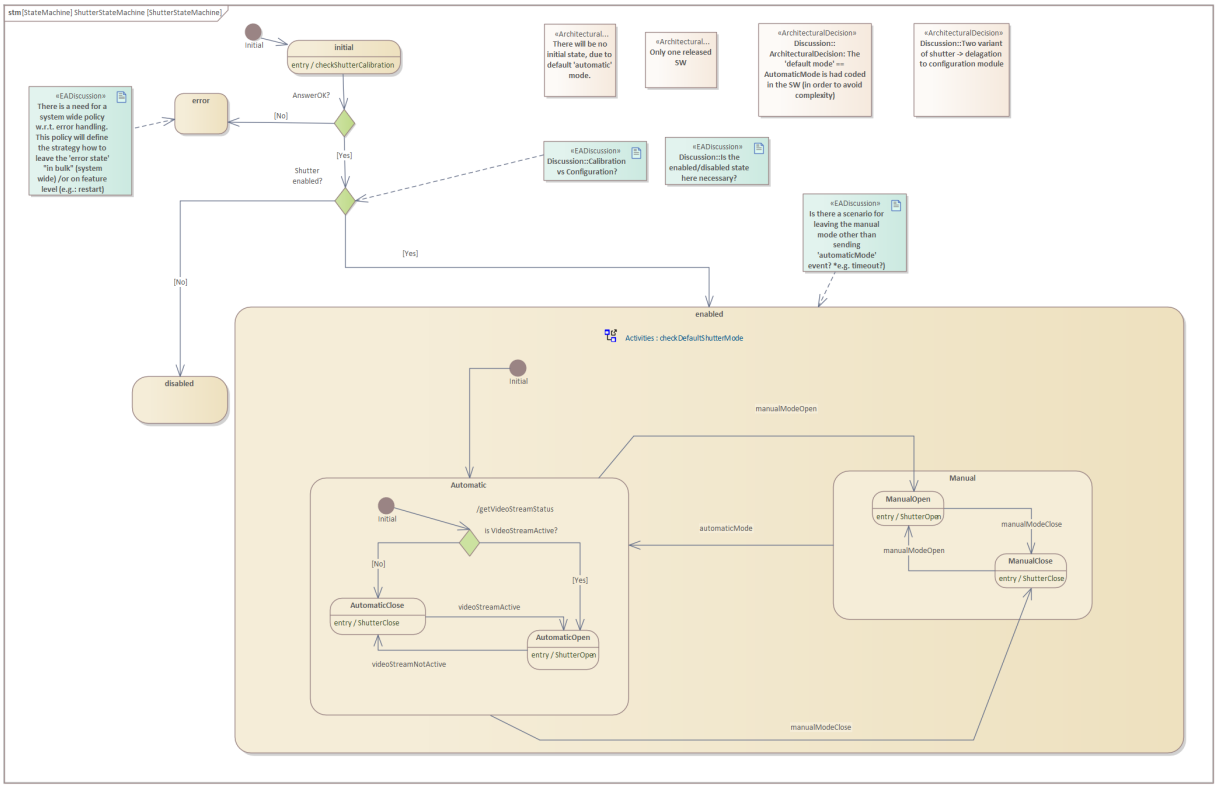


4.3 Shutter

MVC-100996 - Structural

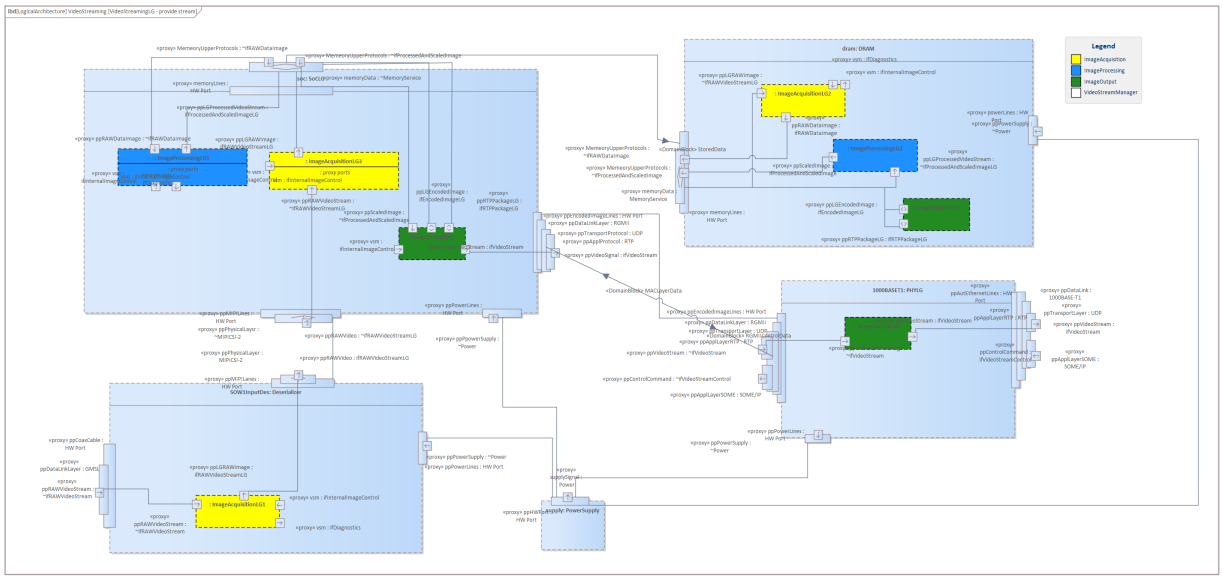


MVC-101010 - Behavioral

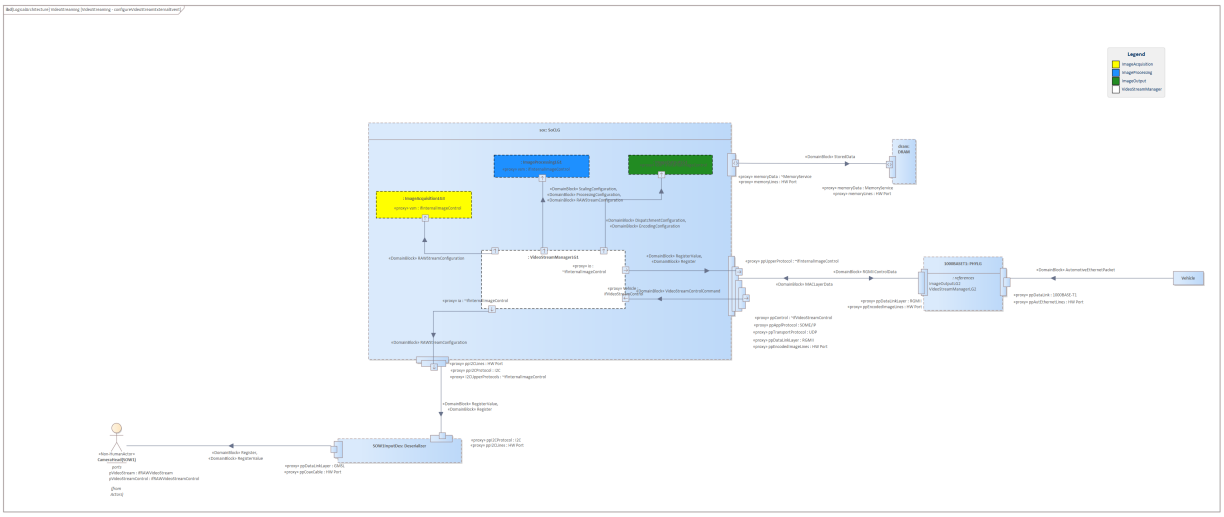


4.4 Video Streaming

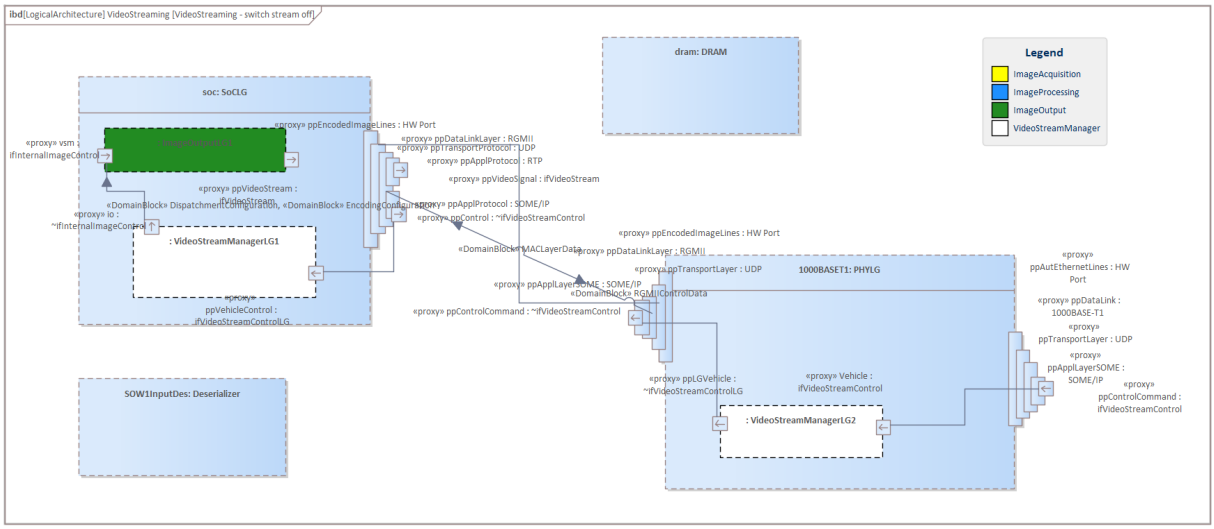
MVC-100997 - Structural - ProvideStream



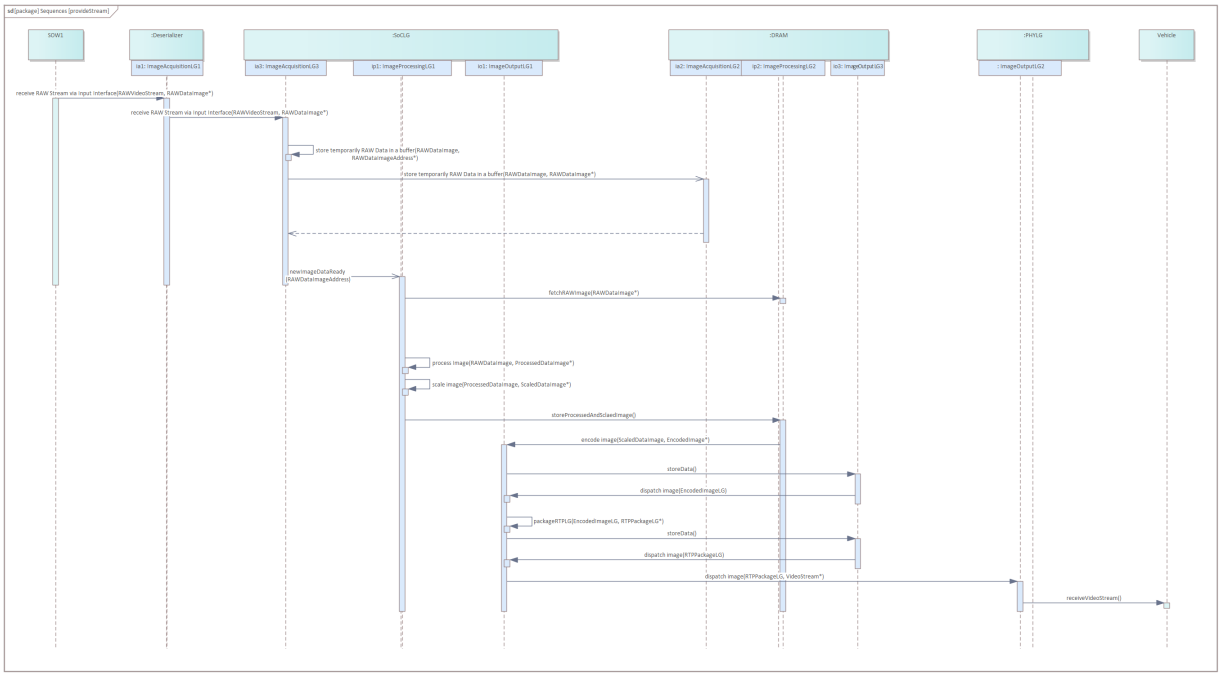
MVC-101015 - Structural - ConfigureVideoStreamExternalEvent

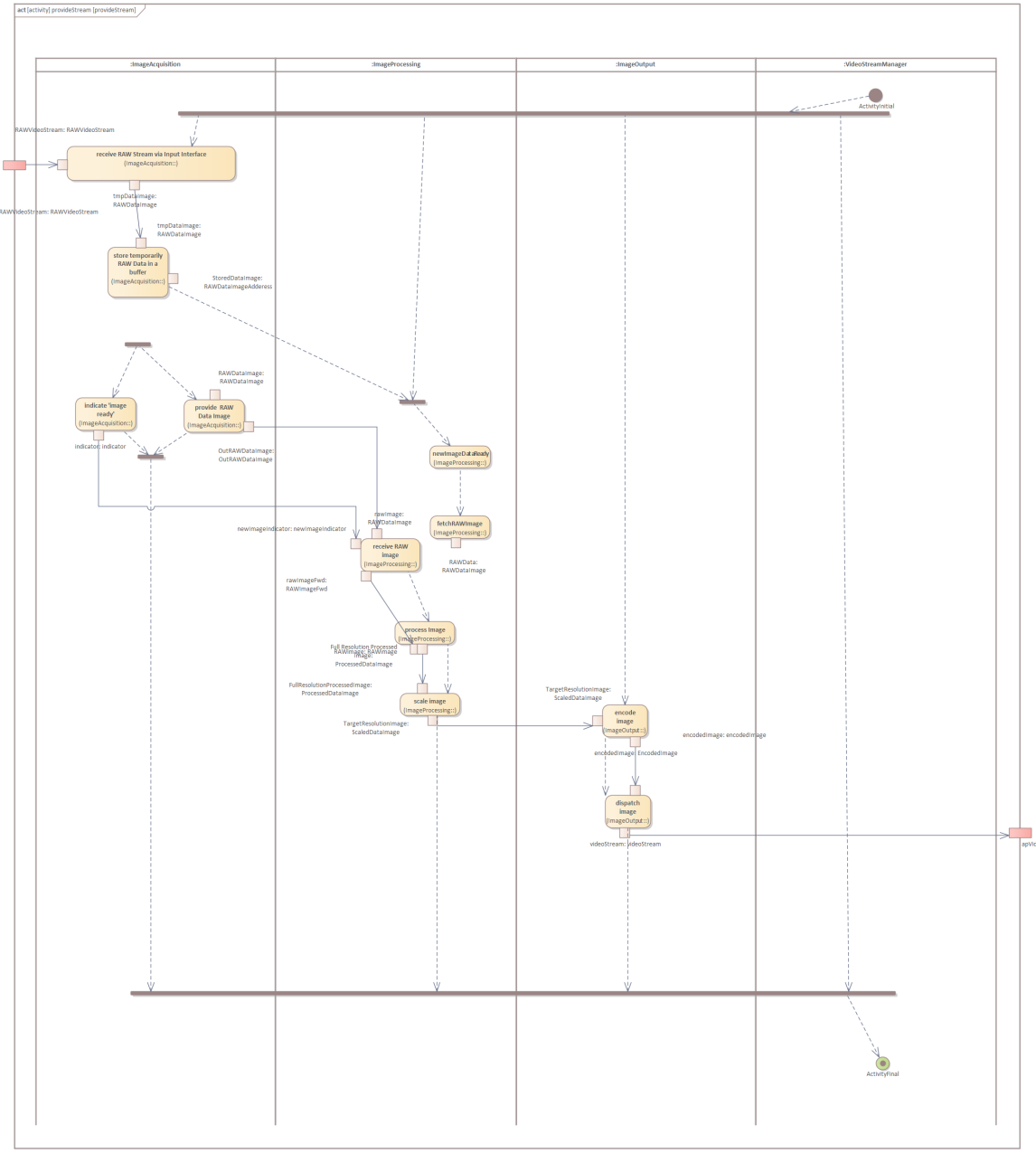


MVC-101016 - Structural - SwitchStreamOff

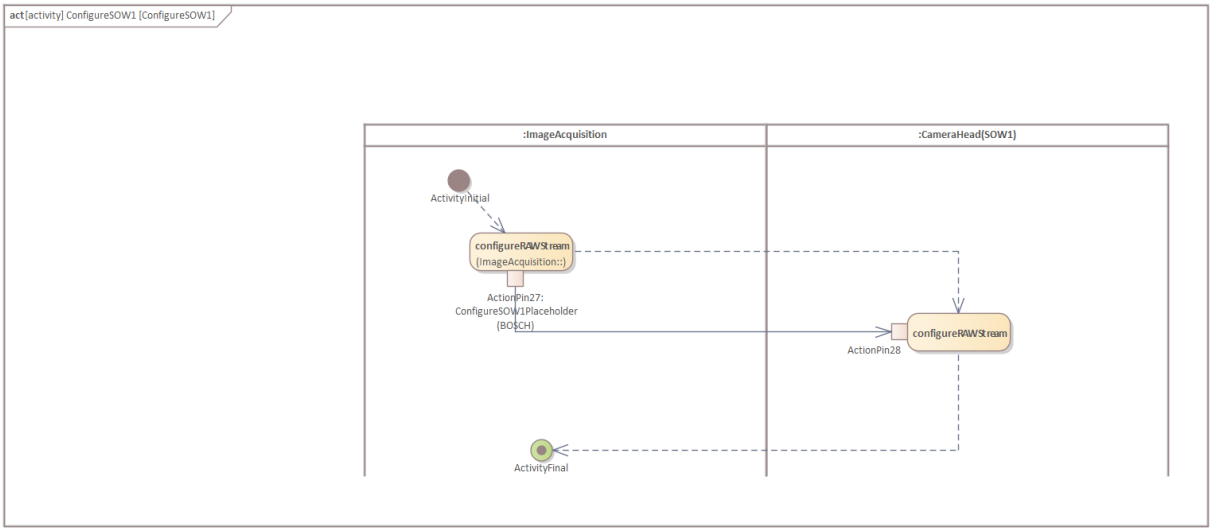


MVC-101011 - Behavioral - ProvideStream

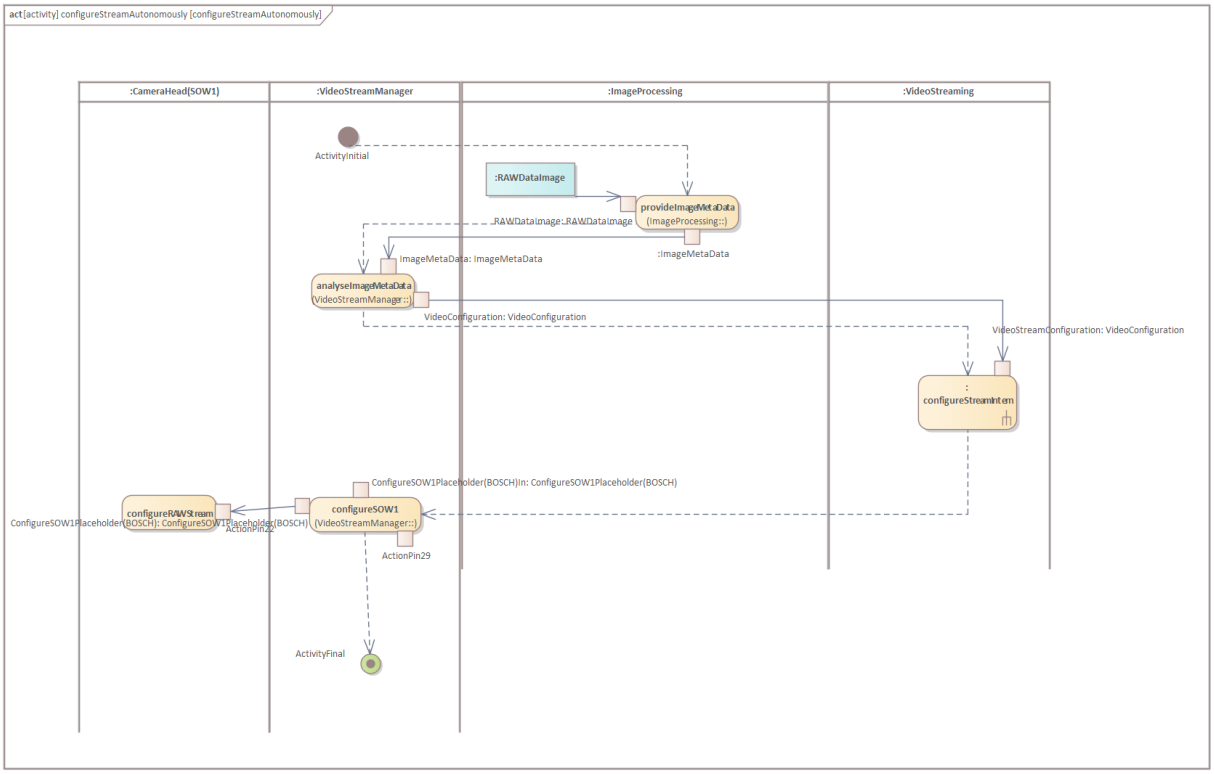

sd [package] Sequences [providedStream]
SDW1
:Deserializer
ia1: ImageAcquisition:LG1
:SoC:LG
ia3: ImageAcquisition:LG3
ip1: ImageProcessing:LG1
io1: ImageOutput:LG1
:DRAM
ia2: ImageAcquisition:LG2
ip2: ImageProcessing:LG2
io3: ImageOutput:LG3
:PHY:LG
: ImageOutput:LG2
Vehicle
receive RAW Stream via input interface(RAWVideoStream, RAWDataImage*)
receive RAW Stream via input interface(RAWVideoStream, RAWDataImage*)
store temporarily RAW Data in a buffer(RAWDataImage, RAWDataImageAddress*)
store temporarily RAW Data in a buffer(RAWDataImage, RAWDataImage*)
newImageDataReady(RAWDataImageAddress)
fetchRAWImage(RAWDataImage*)
process Image(RAWDataImage, ProcessedDataImage*)
scale image(ProcessedDataImage, ScaledDataImage*)
storeProcessedAndScaledImage()
encode image(ScaledDataImage, EncodedImage*)
storeData()
dispatch image(EncodedImage:LG)
packageRTP:LG(EncodedImage:LG, RTPPackage:LG*)
storeData()
dispatch image(RTPPackage:LG)
dispatch image(RTPPackage:LG, VideoStream*)
receiveVideoStream()



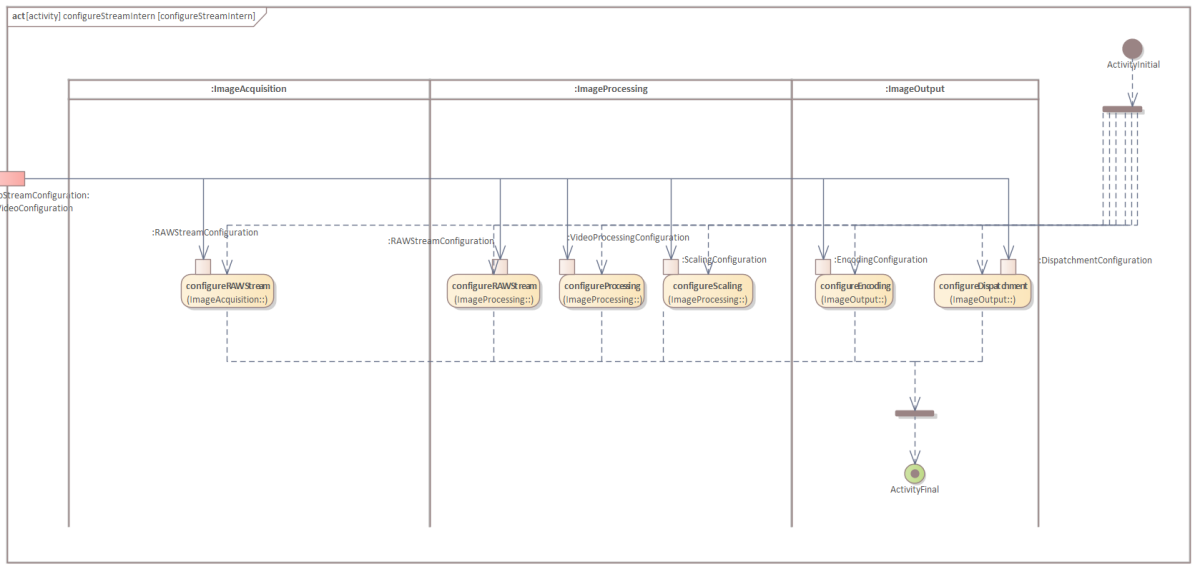
MVC-101022 - Behavioral - ConfigureSOW1



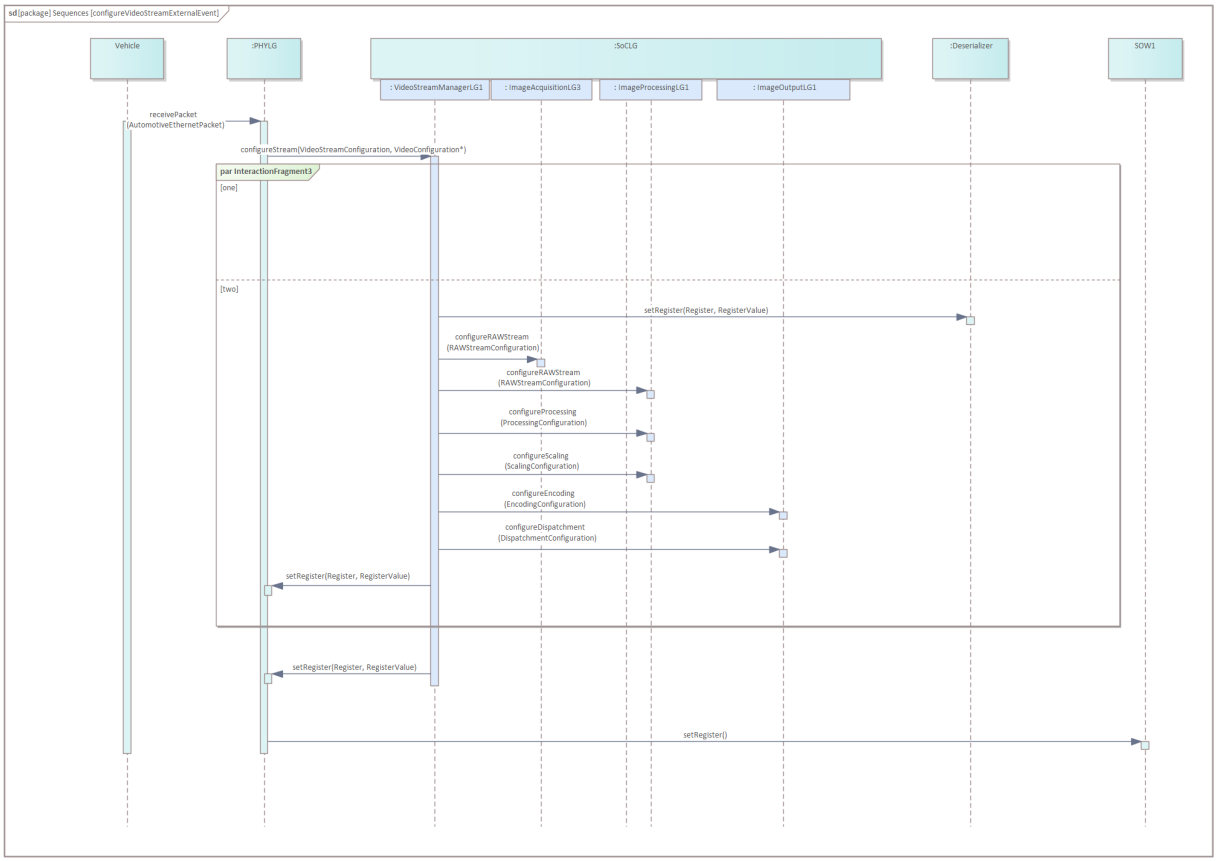
MVC-101023 - Behavioral - ConfigureStreamAutonomously

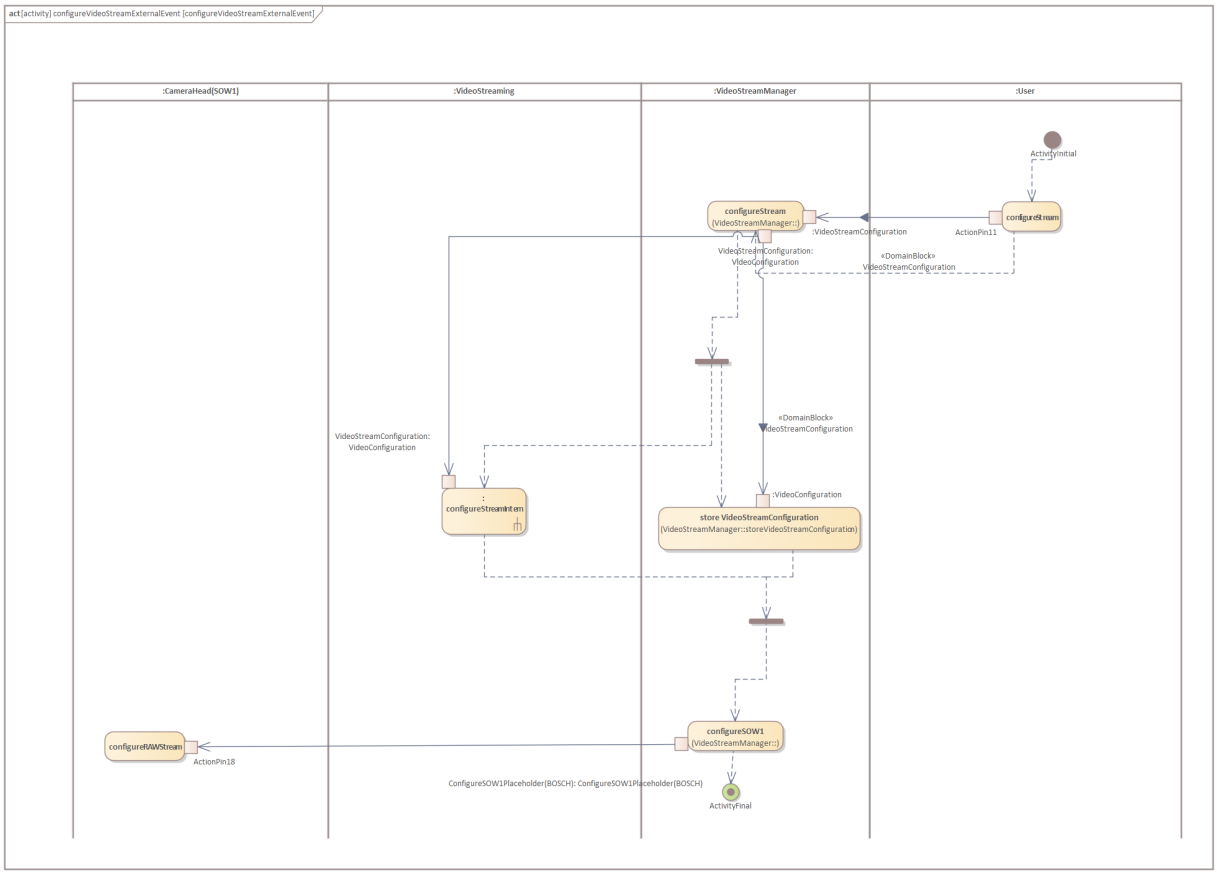


MVC-101024 - Behavioral - ConfigureStreamIntern

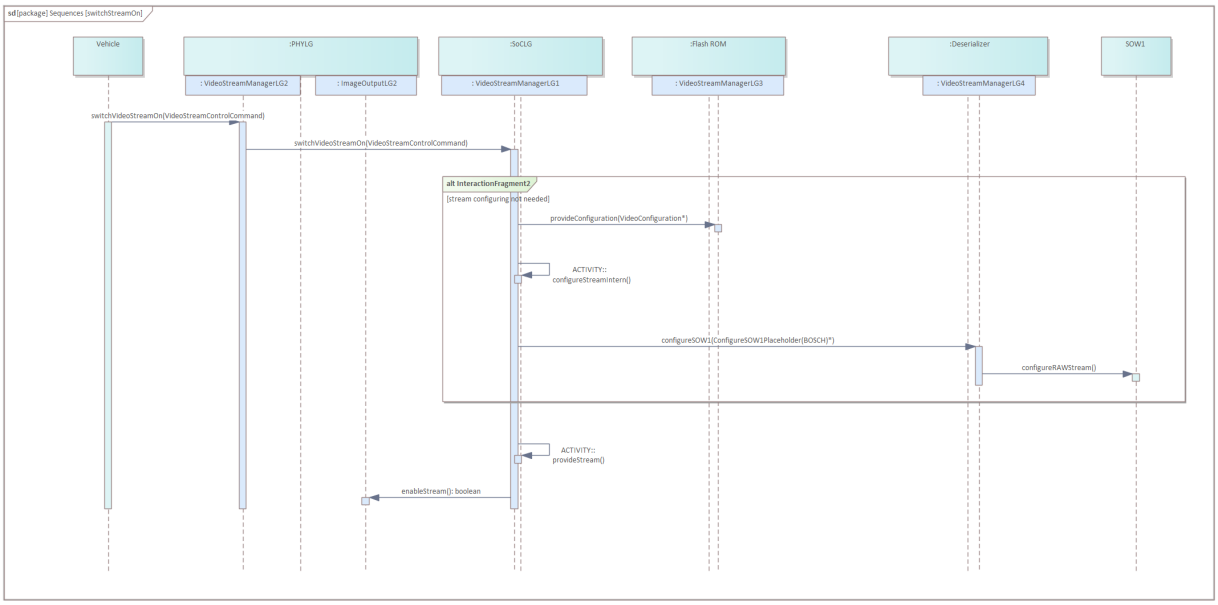


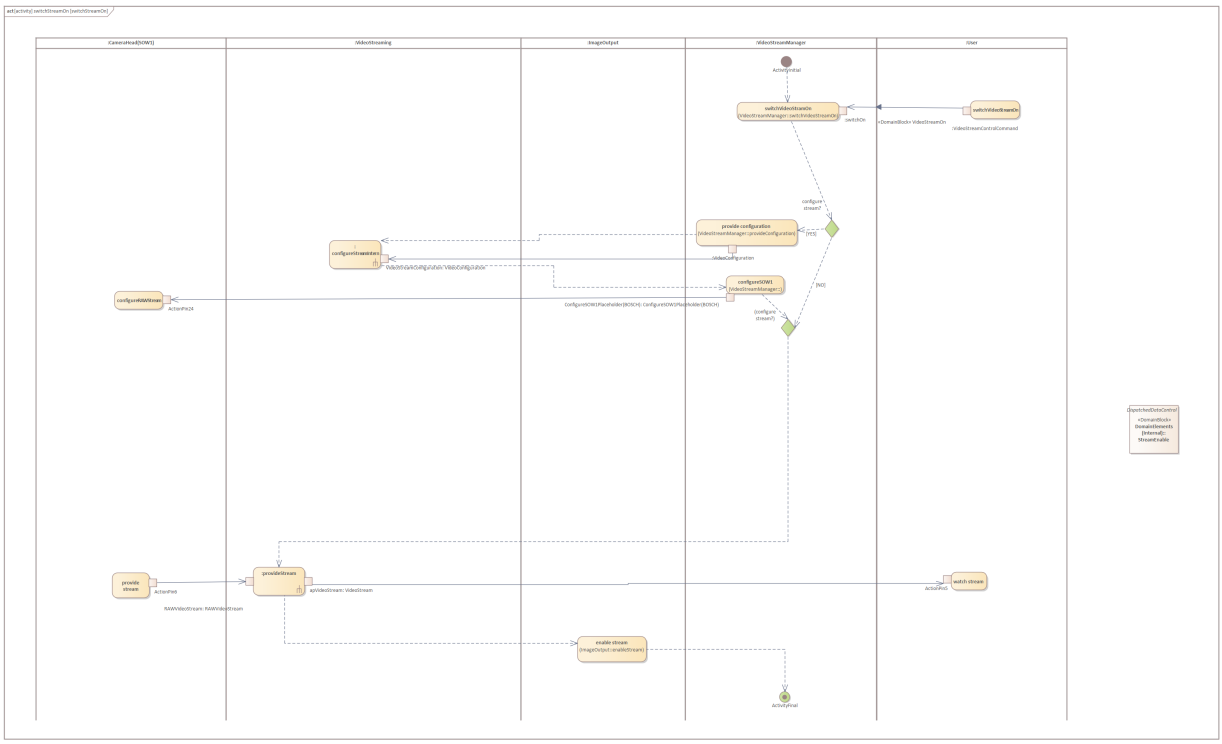
MVC-101025 - Behavioral - ConfigureVideoStreamExternalEvent



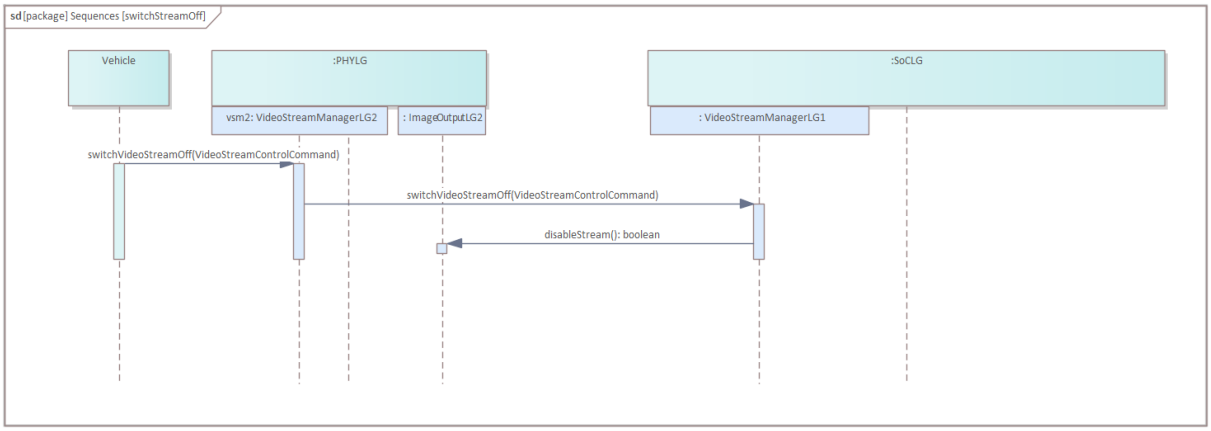


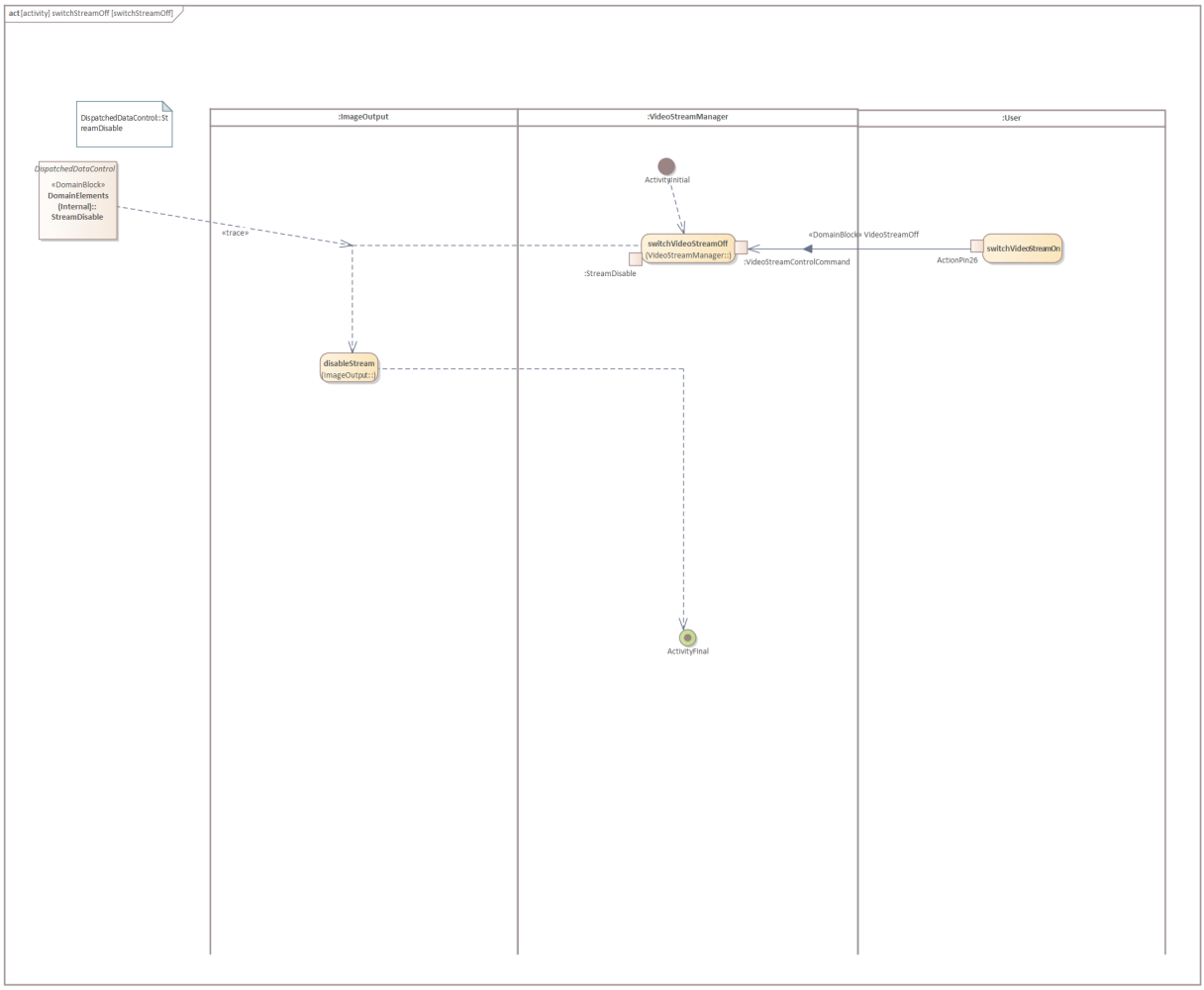
MVC-101026 - Behavioral - SwitchStreamOn





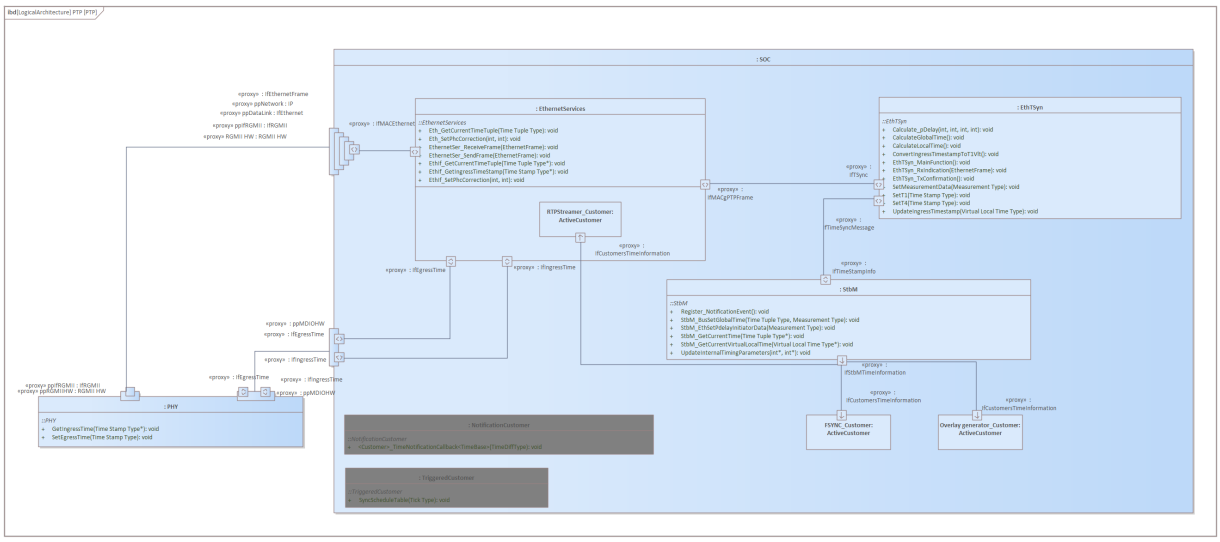
MVC-101027 - Behavioral - SwitchStreamOff



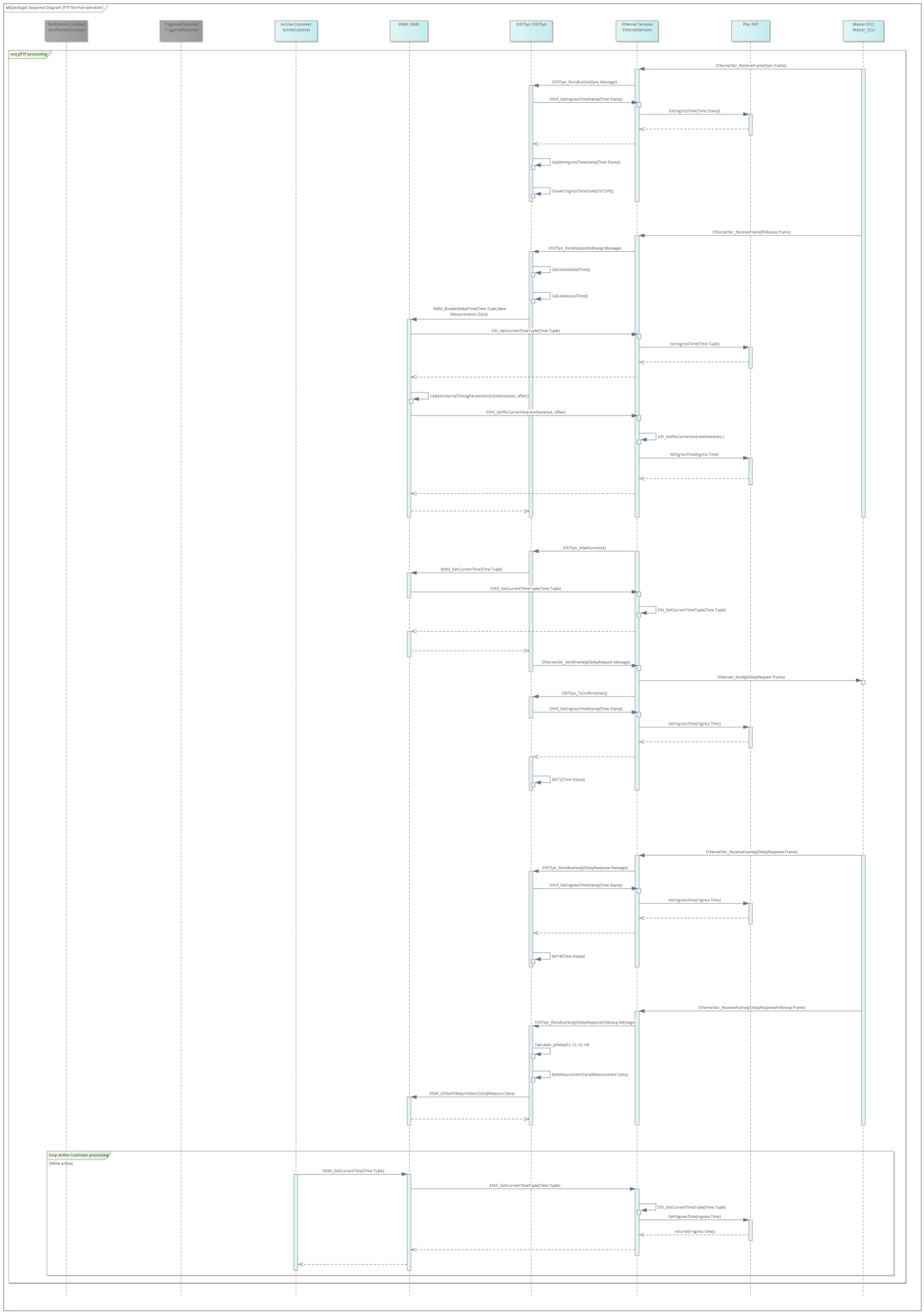


4.5 PTP

MVC-100998 - Structural

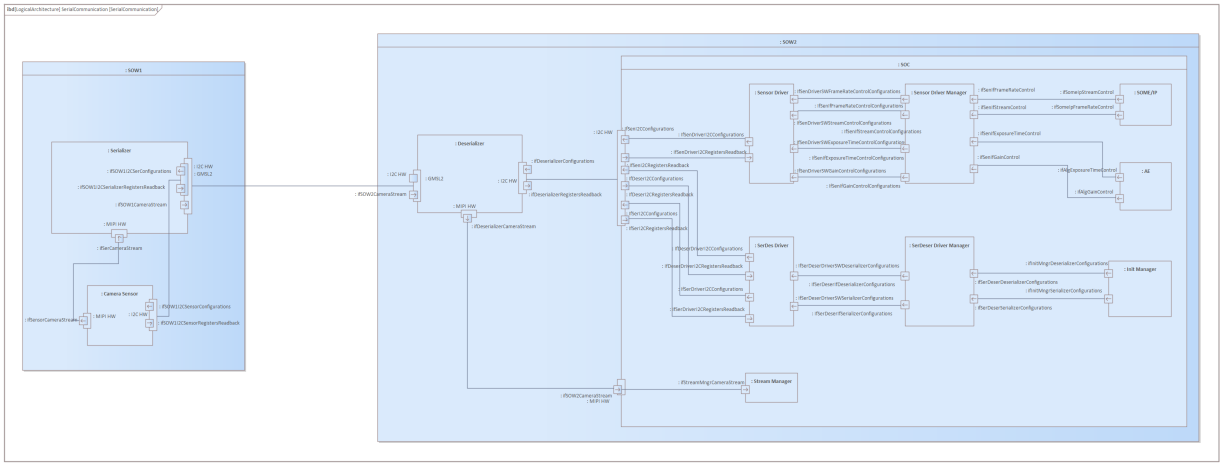


MVC-101012 - Behavioral


sd [package] Sequence Diagram [PTP Normal operation]
NotificationCustomer: NotificationCustomer
TriggeredCustomer: TriggeredCustomer
ActiveCustomer: ActiveCustomer
StbM: StbM
EthTSyn: EthTSyn
EthernetServices: EthernetServices
Phy: PHY
Master ECU: Master_ECU
seq gPTP processing
EthernetSer_ReceiveFrame(Sync Frame)
EthTSyn_RxIndication(Sync Message)
EthIf_GetIngressTimeStamp(Time Stamp)
GetIngressTime(Time Stamp)
UpdateIngressTimestamp(Time Stamp)
ConvertIngressTimestampToT1vItl()
EthernetSer_ReceiveFrame(Followup Frame)
EthTSyn_RxIndication(Followup Message)
CalculateGlobalTime()
CalculateLocalTime()
StbM_BusSetGlobalTime(Time Tuple,New Measurements Data)
Eth_GetCurrentTimeTuple(Time Tuple)
GetIngressTime(Time Tuple)
UpdateInternalTimingParameters(rateDeviation, offset)
EthIf_SetPhcCorrection(rateDeviation, offset)
Eth_SetPhcCorrection(rateDeviation,)
SetEgressTime(Egress Time)
EthTSyn_MainFunction()
StbM_GetCurrentTime(Time Tuple)
EthIf_GetCurrentTimeTuple(Time Tuple)
Eth_GetCurrentTimeTuple(Time Tuple)
EthernetSer_SendFrame(pDelayRequest Message)
Ethernet_Send(pDelayRequest Frame)
EthTSyn_TxConfirmation()
EthIf_GetIngressTimeStamp(Time Stamp)
GetIngressTime(Ingress Time)
SetT3(Time Stamp)
EthernetSer_ReceiveFrame(pDelayResponse Frame)
EthTSyn_RxIndication(pDelayResponse Message)
EthIf_GetIngressTimeStamp(Time Stamp)
GetIngressTime(Ingress Time)
SetT4(Time Stamp)
EthernetSer_ReceiveFrame(pDelayResponseFollowup Frame)
EthTSyn_RxIndication(pDelayResponseFollowup Message)
Calculate_pDelay(t1, t2, t3, t4)
SetMeasurementData(Measurement Data)
StbM_EthSetPdelayInitiatorData(Measure Data)
loop Active Customer processing
[While active]
StbM_GetCurrentTime(Time Tuple)
EthIf_GetCurrentTimeTuple(Time Tuple)
Eth_GetCurrentTimeTuple(Time Tuple)
GetIngressTime(Ingress Time)
returned Ingress time()

4.6 Camera Interface

MVC-100999 - Structural

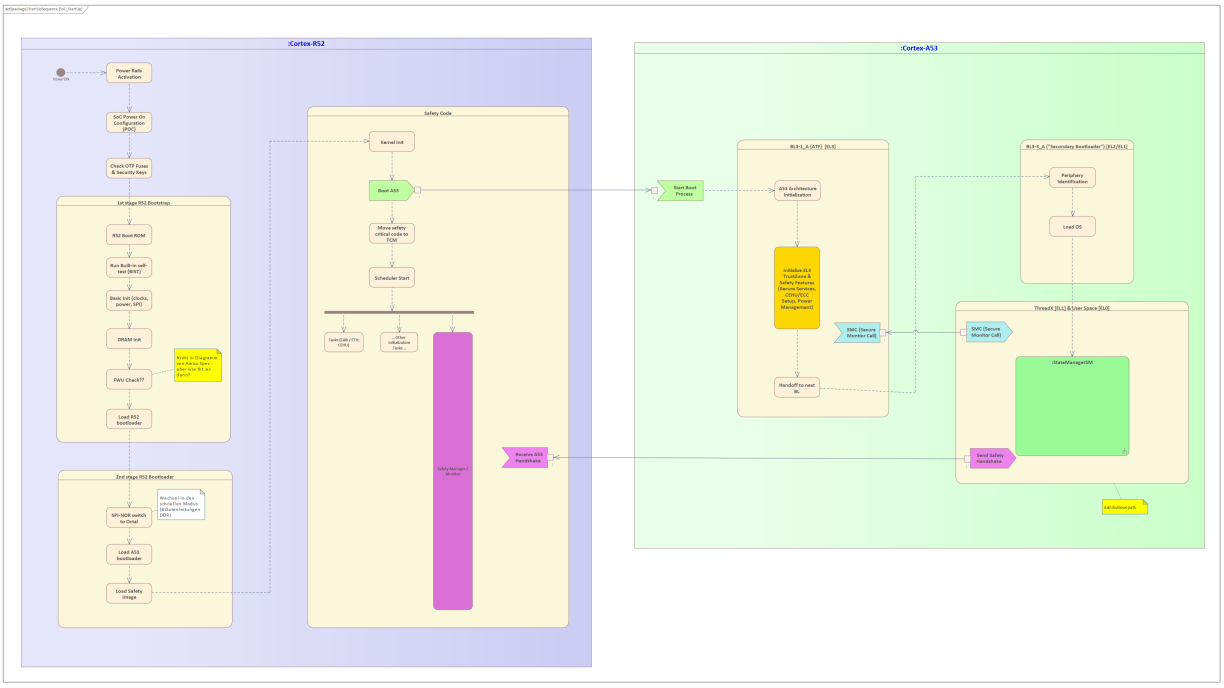


MVC-101013 - Behavioral

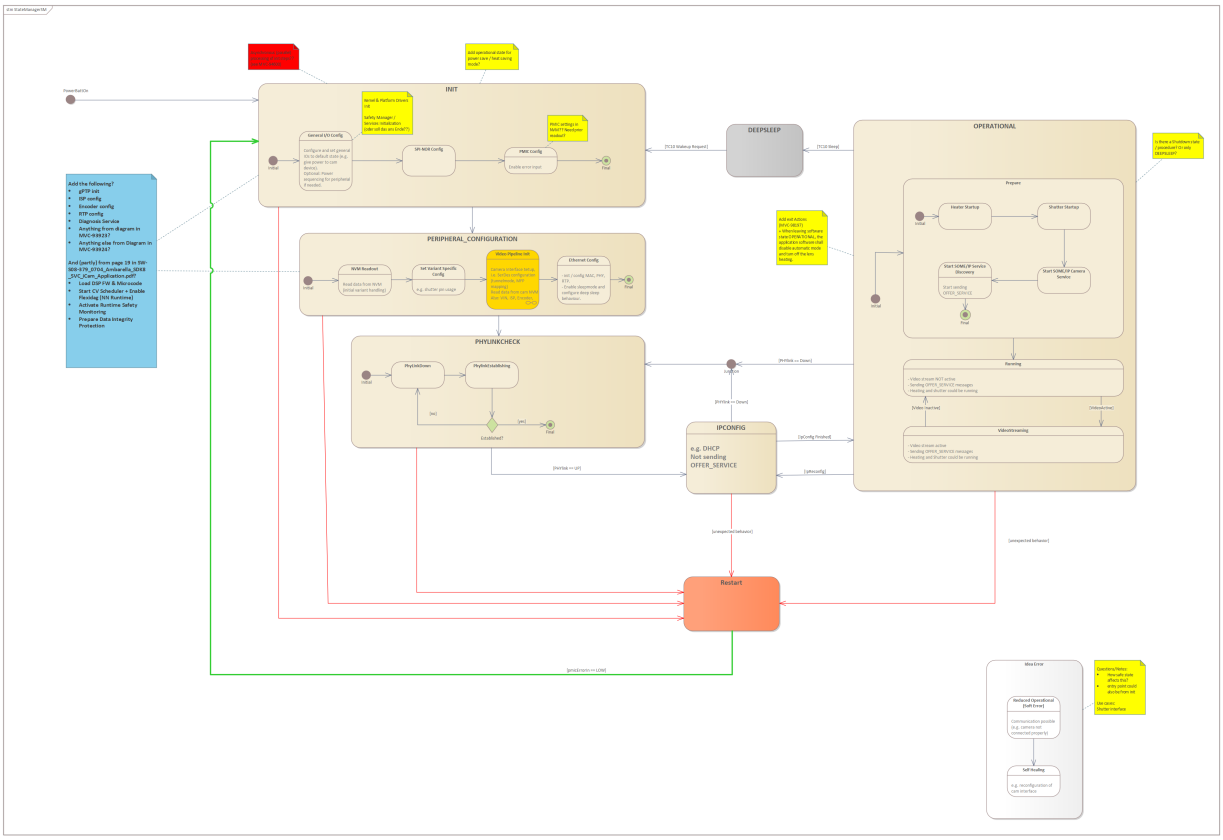
4.7 State Handling

MVC-101000 - Structural

MVC-101014 - Behavioral - SoC Startup



MVC-101028 - Behavioral - A-Core Lifecycle States


INIT
DEEPSLEEP
OPERATIONAL
PERIPHERAL_CONFIGURATION
PHYLINKCHECK
IPCONFIG
e.g. DHCP
Not sending
OFFER_SERVICE
Restart